

Part II static projective compactness

A fixed-coarse interacting scaling state along a cofinal subnet

Lamport-form proof

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Unconditional static conclusion. Once one state has been chosen on every finite-resolution algebra, weak-star compactness produces a cofinal subnet whose pullbacks converge simultaneously on every fixed coarse algebra. The limiting family is projectively consistent and therefore defines a state on the C^* -inductive limit.

The theorem proves neither convergence of the full sequence nor identification with the continuum $P(\phi)_2$ vacuum. It gives no rate, no local-state-norm convergence, no normality, and no limiting dynamics.

Contents

1	Statement	1
2	Proof of Theorem 1.2	2
3	Spatial-cutoff interface	3
4	Preserved compatible symmetries	4
5	Sharp limitations	4

1 Statement

Definition 1.1 (directed algebraic system). For $M \leq N$, let $\mathfrak{A}_M, \mathfrak{A}_N$ be unital C^* -algebras and let

$$\alpha_M^N : \mathfrak{A}_M \longrightarrow \mathfrak{A}_N \tag{1.1}$$

be injective unital $*$ -homomorphisms such that

$$\alpha_N^K \circ \alpha_M^N = \alpha_M^K, \quad M \leq N \leq K. \tag{1.2}$$

Let $(\mathfrak{A}_\infty, \alpha_M^\infty)$ be their C^* -inductive limit, so

$$\alpha_N^\infty \circ \alpha_M^N = \alpha_M^\infty, \quad \overline{\bigcup_M \alpha_M^\infty(\mathfrak{A}_M)}^{\|\cdot\|} = \mathfrak{A}_\infty. \tag{1.3}$$

Theorem 1.2 (cofinal-subnet projective limit). *For every N , let ω_N be a state on $\mathfrak{A}_N$. Then there exist a directed set I , a cofinal subnet map $i \mapsto N(i) \in \mathbb{N}_0$, states $\rho_M \in S(\mathfrak{A}_M)$, and a state*

$\Omega_\infty \in S(\mathfrak{A}_\infty)$ such that

$$\lim_{i \in I} \omega_{N(i)}(\alpha_M^{N(i)}(A)) = \rho_M(A) = \Omega_\infty(\alpha_M^\infty(A)), \quad (1.4)$$

$$\rho_M = \rho_{M'} \circ \alpha_M^{M'}, \quad M \leq M', \quad (1.5)$$

for every fixed M and every $A \in \mathfrak{A}_M$.

2 Proof of Theorem 1.2

⟨1⟩ 1 Prove weak-star compactness at one scale.

The closed unit ball of $\mathfrak{A}_M^*$ is compact in $\sigma(\mathfrak{A}_M^*, \mathfrak{A}_M)$ by Banach–Alaoglu. Within that ball,

$$S(\mathfrak{A}_M) = \{\varphi : \varphi(1) = 1, \varphi(A^*A) \geq 0 \text{ for every } A \in \mathfrak{A}_M\}. \quad (2.1)$$

For fixed A , both maps

$$\varphi \mapsto \varphi(1), \quad \varphi \mapsto \varphi(A^*A) \quad (2.2)$$

are weak-star continuous. Hence every condition in (2.1) is weak-star closed. Therefore $S(\mathfrak{A}_M)$ is weak-star compact.

⟨1⟩ 2 Put all coarse restrictions into one compact space.

Define

$$K := \prod_{M=0}^{\infty} S(\mathfrak{A}_M), \quad (2.3)$$

with the product of the weak-star topologies. Tychonoff’s theorem makes K compact. For every N , define $x_N = (x_N^M)_{M \geq 0} \in K$ by

$$x_N^M := \begin{cases} \omega_N \circ \alpha_M^N, & M \leq N, \\ \omega_M, & M > N. \end{cases} \quad (2.4)$$

The second case only fills coordinates that are eventually irrelevant.

⟨1⟩ 3 Extract a genuinely cofinal subnet.

Every net in a compact space has a convergent subnet. Hence there are a directed set I , a subnet map $N : I \rightarrow \mathbb{N}_0$, and $(\rho_M)_M \in K$ such that

$$x_{N(i)} \longrightarrow (\rho_M)_M \text{ in } K. \quad (2.5)$$

By the definition of a subnet, the map $i \mapsto N(i)$ is cofinal: for every M' , there is i_0 such that $N(i) \geq M'$ whenever $i \geq i_0$. Product convergence in (2.5) means, explicitly,

$$\lim_i x_{N(i)}^M(A) = \rho_M(A), \quad A \in \mathfrak{A}_M. \quad (2.6)$$

⟨1⟩ 4 Prove projective consistency.

Fix $M \leq M'$ and $A \in \mathfrak{A}_M$. Cofinality implies that $N(i) \geq M'$ eventually. For those i , (1.2) gives

$$x_{N(i)}^{M'}(\alpha_M^{M'}(A)) = \omega_{N(i)}(\alpha_{M'}^{N(i)}(\alpha_M^{M'}(A))) \quad (1)$$

$$= \omega_{N(i)}(\alpha_M^{N(i)}(A)) = x_{N(i)}^M(A). \quad (2.7)$$

Take the limit in (2.7) and use (2.6):

$$\rho_{M'}(\alpha_M^{M'}(A)) = \rho_M(A). \quad (2.8)$$

This is exactly (1.5).

⟨1⟩ 5 Define the functional on the algebraic inductive union.

Set

$$\mathfrak{A}_{\text{alg}} := \bigcup_M \alpha_M^\infty(\mathfrak{A}_M) \quad (2.9)$$

and prescribe

$$\Omega_0(\alpha_M^\infty(A)) := \rho_M(A). \quad (2.10)$$

To prove that (2.10) is well-defined, suppose

$$\alpha_M^\infty(A) = \alpha_{M'}^\infty(B). \quad (2.11)$$

Choose $K \geq M, M'$. By (1.3) and injectivity of α_K^∞ ,

$$\alpha_M^K(A) = \alpha_{M'}^K(B). \quad (2.12)$$

Applying ρ_K and then (2.8) gives

$$\rho_M(A) = \rho_K(\alpha_M^K(A)) = \rho_K(\alpha_{M'}^K(B)) = \rho_{M'}(B). \quad (2.13)$$

Thus (2.10) is independent of the representative.

⟨1⟩ 6 Prove positivity, normalization, and boundedness.

Every finite collection of elements of $\mathfrak{A}_{\text{alg}}$ lies in the image of one $\mathfrak{A}_K$. Consequently linearity of Ω_0 follows from linearity of ρ_K . If $C = \alpha_M^\infty(A)$, then

$$\Omega_0(C^*C) = \rho_M(A^*A) \geq 0, \quad \Omega_0(1) = 1. \quad (2.14)$$

For completeness, positivity implies the Cauchy–Schwarz inequality

$$|\rho_M(A)|^2 \leq \rho_M(A^*A)\rho_M(1) \leq \|A\|^2. \quad (2.15)$$

Because every injective $*$ -homomorphism of C^* -algebras is isometric,

$$|\Omega_0(\alpha_M^\infty(A))| \leq \|A\| = \|\alpha_M^\infty(A)\|. \quad (2.16)$$

Thus Ω_0 is bounded with norm one.

⟨1⟩ 7 Extend and identify the coordinate limits.

By (1.3), $\mathfrak{A}_{\text{alg}}$ is norm dense in $\mathfrak{A}_\infty$. Therefore Ω_0 extends uniquely and continuously to a state Ω_∞ on $\mathfrak{A}_\infty$. Equations (2.6) and (2.10) give (1.4). $\square$

3 Spatial-cutoff interface

Corollary 3.1 (iterated cutoff then resolution limit). Suppose that for each N there is a directed cutoff family of states $\{\omega_{N,g}\}_g \subset S(\mathfrak{A}_N)$ satisfying

$$\lim_{g \uparrow 1} \omega_{N,g}(B) = \omega_N(B), \quad B \in \mathfrak{A}_N. \quad (3.1)$$

Then the subnet obtained in Theorem 1.2 satisfies

$$\lim_i \left(\lim_{g \uparrow 1} \omega_{N(i),g}(\alpha_M^{N(i)}(A)) \right) = \Omega_\infty(\alpha_M^\infty(A)) \quad (3.2)$$

for every fixed M and $A \in \mathfrak{A}_M$.

Proof. For fixed i , (3.1), applied to $B = \alpha_M^{N(i)}(A)$, makes the inner limit equal to $\omega_{N(i)}(\alpha_M^{N(i)}(A))$. The outer limit is (1.4). $\square$

Remark 3.2 (no joint-limit inference). Equation (3.2) is an iterated limit. It does not imply a joint limit. The scalar array

$$a_{N,R} := \mathbf{1}_{\{R < N\}} \quad (3.3)$$

satisfies $\lim_{R \rightarrow \infty} a_{N,R} = 0$ for every fixed N , but the cofinal diagonal $R = N - 1$ has $a_{N,N-1} = 1$.

4 Preserved compatible symmetries

Proposition 4.1. *Suppose $\vartheta_N \in \text{Aut}(\mathfrak{A}_N)$ obey*

$$\vartheta_N \alpha_M^N = \alpha_M^N \vartheta_M \quad (4.1)$$

and $\omega_N \vartheta_N = \omega_N$ for every N . Then there is an induced $\vartheta_\infty \in \text{Aut}(\mathfrak{A}_\infty)$, and the state from Theorem 1.2 satisfies

$$\Omega_\infty \vartheta_\infty = \Omega_\infty. \quad (4.2)$$

Proof. Fix M, A . Eventually $N(i) \geq M$, and

$$\omega_{N(i)}(\alpha_M^{N(i)}(\vartheta_M A)) = \omega_{N(i)}(\vartheta_{N(i)}(\alpha_M^{N(i)} A)) = \omega_{N(i)}(\alpha_M^{N(i)} A). \quad (4.3)$$

Taking the limit gives $\rho_M \vartheta_M = \rho_M$. Equation (4.2) first follows on $\mathfrak{A}_{\text{alg}}$ and then on $\mathfrak{A}_\infty$ by norm continuity. $\square$

5 Sharp limitations

Proposition 5.1 (the compactness conclusion cannot be upgraded formally). *The hypotheses of Theorem 1.2 imply none of the following:*

1. *convergence of the full sequence;*
2. *convergence in the norm of state functionals;*
3. *normality of the limiting state in a prescribed representation;*
4. *convergence of dynamics or the ground-state spectral condition.*

Proof. For (1), take $\mathfrak{A}_N = \mathbb{C}^2$, all maps equal to the identity, and alternate the two point-evaluation states.

For (2)–(3), take the constant system $\mathfrak{A}_N = B(\ell^2(\mathbb{N}))$ and $\omega_n(A) = \langle e_n, A e_n \rangle$. Every weak-star cluster state ρ annihilates compact operators because $\langle e_n, K e_n \rangle \rightarrow 0$ for every compact K . Hence ρ is singular. For $P_n = |e_n\rangle\langle e_n|$ and $U_n = 2P_n - I$,

$$\|U_n\| = 1, \quad \omega_n(U_n) = 1, \quad \rho(U_n) = -1. \quad (5.1)$$

Therefore $\|\omega_n - \rho\| = 2$, so no norm convergence occurs.

For (4), the hypotheses mention neither automorphism groups nor Hamiltonians. Consequently no statement involving them is a logical consequence. More concretely, one may put arbitrary, mutually incompatible dynamics on the constant algebra without changing any hypothesis or any step of the proof. $\square$

Remark 5.2 (subnet versus subsequence). No separability assumption was made. Thus weak-star compactness yields a subnet, not necessarily a subsequence. If every $\mathfrak{A}_M$ is separable, the state spaces are weak-star metrizable on bounded sets. Choose successively a convergent subsequence at levels $M = 0, 1, 2, \dots$, each contained in the preceding one, and take the diagonal sequence whose j th term is the j th term of the level- j subsequence. For every fixed M its tail lies in the level- M subsequence, so it converges on $\mathfrak{A}_M$; the consistency proof above then applies unchanged.